

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456360.942 +/- 0.06 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.187 +/- 0.055
Transit depth (Rp/Rs)^2	3.5 +/- 2.07 [%]
Orbital Inclination (inc)	86.49 +/- 2.1
Airmass coefficient 1 (a1)	1.32 +/- 0.81
Airmass coefficient 2 (a2)	0.02 +/- 0.16
Scatter in the residuals of the lightcurve fit is	60.99 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	7.26
Transit Duration (day)	0.047 +/- 0.043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.187 ± 0.055 vs 0.1457 (deviation 0.66σ)
Duration fit vs published	0.047 d vs 0.12 d (deviation 1.49σ)
Mid-time O–C	139.8 min
Depth SNR	3.0
Residual scatter	60.99 %

Light curve

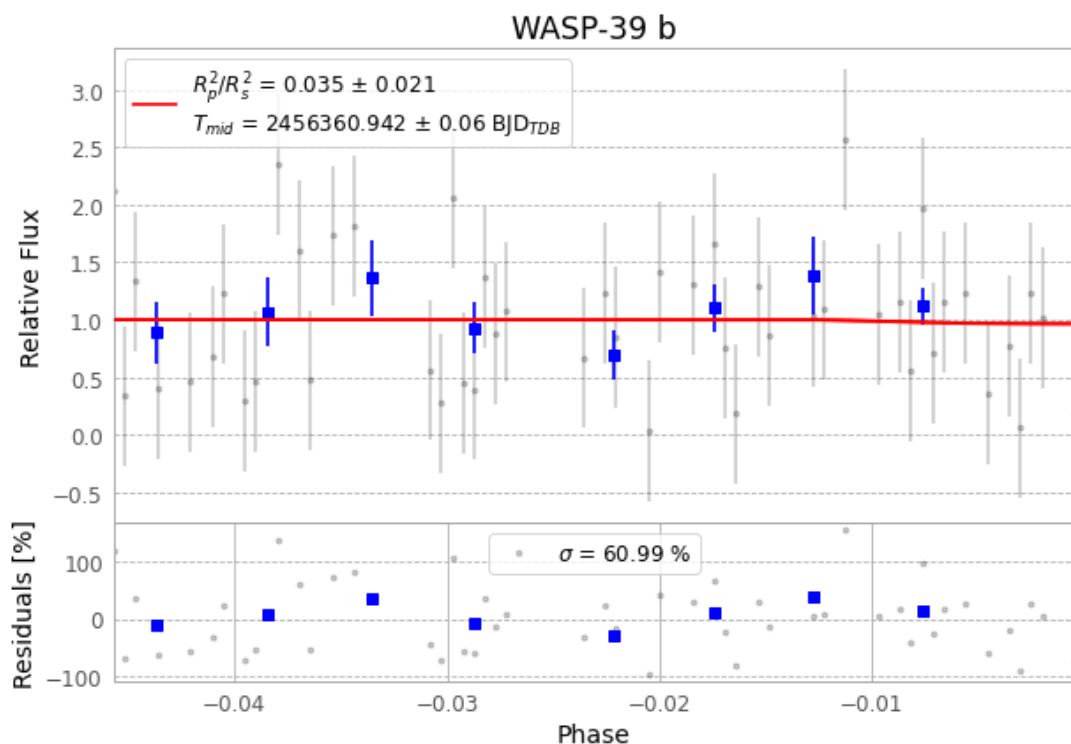


Figure 1 — FinalLightCurve_WASP-39 b_2013-03-08.png (SHA-256 3bc8a27f625e179d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [198, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c2b17c5afe2b0c31...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

91 FITS frames (60 MB), WASP-39130309060312.FITS → WASP-39130309103313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-130309.log (4c04a2168c24...), FinalLightCurve_WASP-39 b_2013-03-08.png (3bc8a27f625e...), FinalLightCurve_WASP-39 b_2013-03-08.csv (84383b94056f...)

Compute resources

Wall time 195.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 00:17Z.